

THOMSEN RIVER, NT, Canada

WMO: 719160

Lat: 73.2311N

Lon: 119.5406W

Elev: 136

StdP: 14.62

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-48.5	-45.7	-57.4	0.2	-48.4	-54.0	0.2	-45.5	35.3	-12.8	31.4	-12.4	2.2	150	1.116

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	66.0	55.6	61.5	53.4	57.0	50.9	56.6	64.8	53.9	60.5	51.2	56.5	12.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.2	58.2	59.6	49.7	53.2	56.7	47.3	48.5	54.0	24.2	64.8	22.6	60.8	21.0	56.6	66.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.8	27.3	24.1	-51.3	70.2	4.1	5.6	-54.2	74.2	-56.6	77.5	-58.9	80.6	-61.9	84.7		
(5)			-51.3	59.1	4.0	3.5	-54.2	61.6	-56.6	63.7	-58.9	65.7	-61.8	68.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-25.0	-25.5	-21.0	-5.2	17.8	39.4	47.5	41.6	29.0	9.3	-8.5	-19.8	(6)
(7)		DBStd	28.51	11.58	12.32	12.60	11.15	9.88	7.74	7.50	7.22	7.87	10.70	11.54	12.51	(7)
(8)		HDD50	15857	2325	2115	2200	1655	999	331	142	278	630	1263	1754	2165	(8)
(9)		HDD65	21238	2790	2535	2665	2105	1464	769	544	724	1080	1728	2204	2630	(9)
(10)		CDD50	95	0	0	0	0	0	12	64	19	0	0	0	0	(10)
(11)		CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)
(12)		CDH74	2	0	0	0	0	0	0	2	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	10.4	9.7	9.0	9.3	10.0	10.8	11.9	12.3	10.9	11.6	10.6	9.6	9.4	(14)
(15)	Precipitation	PrecAvg	5.9	0.2	0.2	0.3	0.3	0.4	0.5	1.1	1.1	0.7	0.6	0.4	0.2	(15)
(16)		PrecMax	8.1	0.6	0.9	0.6	0.8	1.0	1.2	1.9	2.2	1.4	1.5	0.9	1.0	(16)
(17)		PrecMin	3.5	0.0	0.0	0.0	0.0	0.1	0.1	0.4	0.2	0.2	0.1	0.0	0.0	(17)
(18)		PrecStd	1.3	0.1	0.2	0.1	0.2	0.2	0.3	0.4	0.5	0.3	0.3	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	11.8	16.2	30.5	40.0	65.5	72.6	69.9	50.0	33.3	25.5	14.5	(19)
(20)			MCWB	8.9	11.2	15.6	29.5	36.6	55.1	58.0	59.5	45.8	33.2	25.2	14.1	(20)
(21)		2%	DB	1.9	4.4	8.8	21.0	37.2	59.4	67.9	62.9	45.4	30.2	18.2	8.3	(21)
(22)			MCWB	1.5	3.7	8.1	20.0	34.9	52.1	56.1	55.9	43.6	29.8	17.6	7.8	(22)
(23)		5%	DB	-3.2	-0.9	2.7	15.8	34.2	54.8	64.2	56.7	42.4	27.1	12.4	3.2	(23)
(24)			MCWB	-3.6	-1.4	2.1	14.9	32.7	49.0	54.1	52.0	41.4	26.5	11.8	2.6	(24)
(25)		10%	DB	-7.5	-5.8	-2.6	11.8	31.6	50.7	60.2	52.1	39.3	24.3	8.2	-1.3	(25)
(26)			MCWB	-7.9	-6.2	-3.1	11.0	30.1	46.2	52.3	48.6	38.7	23.8	7.6	-1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	11.1	15.8	29.4	37.6	56.3	59.1	60.5	47.1	33.0	25.2	14.2	(27)
(28)			MCDWB	9.1	11.6	16.2	30.5	39.7	63.9	69.4	68.4	48.6	33.0	25.6	14.4	(28)
(29)		2%	WB	1.5	3.9	8.1	20.2	35.2	52.8	57.0	56.5	43.9	29.5	17.6	7.8	(29)
(30)			MCDWB	1.8	4.5	8.7	21.0	36.9	58.5	66.5	62.5	45.3	29.7	18.1	8.3	(30)
(31)		5%	WB	-3.6	-1.3	2.1	15.0	32.7	49.4	54.9	52.3	41.3	26.4	11.9	2.7	(31)
(32)			MCDWB	-3.2	-0.8	2.6	15.7	34.2	54.3	63.0	56.0	42.3	26.8	12.3	3.2	(32)
(33)		10%	WB	-7.9	-6.2	-3.0	11.0	30.3	46.4	52.7	48.8	38.6	23.7	7.7	-1.7	(33)
(34)			MCDWB	-7.5	-5.8	-2.6	11.8	31.5	50.6	59.4	51.7	39.3	24.2	8.2	-1.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.3	12.8	14.3	17.1	12.9	10.3	12.5	10.9	7.8	11.5	13.3	12.8	(35)
(36)			MCDBR	18.9	17.1	15.1	16.8	13.0	15.9	17.0	17.6	10.4	10.4	16.3	16.8	(36)
(37)			MCWBR	18.7	17.0	14.8	15.9	10.7	9.6	8.2	10.7	8.4	10.6	16.2	16.8	(37)
(38)		5% WB	MCDWB	18.9	16.7	15.3	16.4	12.2	14.9	16.8	16.5	9.8	10.7	16.3	16.8	(38)
(39)			MCWBR	18.8	16.6	15.0	15.6	10.1	9.7	9.2	10.3	8.1	10.7	16.3	16.8	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	N/A	0.177	0.244	0.316	0.331	0.334	0.342	0.331	0.280	0.173	N/A	N/A	(41)	
(42)		taud	N/A	1.901	2.152	2.017	2.093	2.374	2.420	2.433	2.397	1.928	N/A	N/A	(42)	
(43)		Ebn at Noon	0	123	226	243	259	267	259	245	218	116	0	0	(43)	
(44)		Edh at Noon	0	8	20	35	38	30	28	24	17	7	0	0	(44)	
(45)	All-Sky Solar Radiation	RadAvg	0	54	463	1312	1921	1863	1574	936	369	99	0	0	(45)	
(46)		RadStd	0	6	24	61	141	175	145	112	38	15	0	0	(46)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page